Literature status: continuous SCP does not imply a projectional skeleton

Source arXiv:1105.4262; answer arXiv:1209.0199

11 August 2026

Status: literature already answered (Question 1.4). The answer is no. There is a Banach space with continuous 2-SCP—indeed, also controlled SCP—which has no projectional skeleton.

Original question

Kalenda and Kubiś ask in Question 1.4 on printed page 3 of [1]:

> Let E be a Banach space with the continuous k-SCP. Does it have a projectional skeleton?

Here continuous k-SCP means that a closed cofinal family of separable subspaces of E consists of k-complemented subspaces.

Separate negative answer

Ferrer, Koszmider, and Kubiś state in Theorem 1.6 of [2] (printed page 6): there is an almost disjoint family $\mathcal{A}$ such that $C(K_{\mathcal{A}})$ has controlled SCP and continuous SCP, but has neither a projectional resolution of the identity nor a projectional skeleton.

The proof in Section 5 (printed pages 17–18) is quantitative enough for the source hypothesis: it writes $C(K_{\mathcal{A}})$ as the union of a continuous chain $(E_\alpha)_{\alpha<\omega_1}$ of separable subspaces and constructs a projection onto every E_α of norm at most 2. Thus this space has continuous 2-SCP while lacking a projectional skeleton, directly falsifying Question 1.4.

Provenance and scope

The supporting paper cites [1], uses its continuous-SCP terminology, and says that its examples decide the missing implications among the separable complementation properties under discussion. This is therefore an explicit later literature answer, not a new counterexample from this run. The identification is limited to Question 1.4: source Question 1.6 about $\aleph_0$-monolithic dual balls and source Conjecture 4.2 for scattered compact lines are not resolved by this packet.

References

[1] O. F. K. Kalenda and W. Kubiś, *Complementation in spaces of continuous functions on compact lines*, J. Math. Anal. Appl. 386 (2012), 241–257; arXiv:1105.4262; DOI: 10.1016/j.jmaa.2011.07.057.

[2] J. Ferrer, P. Koszmider, and W. Kubiś, *Almost disjoint families of countable sets and separable complementation properties*, J. Math. Anal. Appl. 401 (2013), 216–229; arXiv:1209.0199; DOI: 10.1016/j.jmaa.2012.11.065.